

NSF Grant Proposal

Directorate for Biological Sciences

Proposal Number: BIO-2024-72816

Submission Date: November 1, 2024

Project Information

Project Title: Privacy-Preserving Federated Learning for Rural Healthcare Networks

Principal Investigator: Dr. Robert Kim

Institution: University of Idaho

Department: Department of Computer Science

Requested Amount: \$275,000

Project Period: January 1, 2026 - December 31, 2028

Sponsoring Agency: National Science Foundation

Research Area: Computer Science

Project Abstract

Rural healthcare providers often lack sufficient patient data to train accurate machine learning models for clinical decision support. This project develops a federated learning framework that enables multiple rural hospitals and clinics to collaboratively train models without sharing sensitive patient data. We will implement differential privacy guarantees, develop communication-efficient protocols for low-bandwidth rural networks, and validate our approach on three clinical prediction tasks: emergency department triage, readmission risk, and medication interaction detection. The project includes partnerships with 5 critical access hospitals in Idaho.